Popular Arduino Sensors & Input Modules: HC-SR04, DHT11, LDR, PIR, Potentiometer

Document: EDU-SENSOR-01 | Comprehensive Pinout, Wiring & Specification Reference

1. HC-SR04 Ultrasonic Distance Sensor

Operating Voltage: 5V DC. Operating Current: 15mA. Measurement Range: 2cm to 400cm. Resolution: 3mm. Measuring Angle: 15 degrees cone. Trigger Input Signal: 10 microsecond TTL high pulse. Output Echo Signal: TTL level proportional to measured distance.

Pinout (4 pins, left to right): Pin 1 = VCC (+5V), Pin 2 = TRIG (Trigger input, connect to any Arduino digital pin), Pin 3 = ECHO (Echo output, connect to any Arduino digital pin), Pin 4 = GND (Ground). IMPORTANT: The ECHO pin outputs a 5V signal; on a 3.3V microcontroller like ESP32, use a voltage divider (1k + 2k resistors) to step down ECHO to 3.3V or risk damaging the GPIO.

```
Distance Formula: distance_cm = (echo_pulse_duration_microseconds * 0.0343) / 2. Speed of sound at 20C = 343 m/s = 0.0343 cm/us. Division by 2 accounts for the round-trip (sound travels to object and back). Minimum trigger-to-trigger interval: 60 milliseconds to avoid echo overlap.
```

2. DHT11 / DHT22 Temperature & Humidity Sensor

DHT11 Specifications: Operating Voltage 3.3V to 5.5V. Temperature Range: 0 to 50 degrees Celsius (+/- 2C accuracy). Humidity Range: 20% to 80% RH (+/- 5% accuracy). Sampling Rate: Maximum 1 reading per second (1 Hz). Communication: Single-wire bidirectional serial protocol on DATA pin.

DHT22 (AM2302) Specifications: Temperature Range: -40 to 80 degrees Celsius (+/- 0.5C accuracy). Humidity Range: 0% to 100% RH (+/- 2% accuracy). Sampling Rate: Maximum 1 reading every 2 seconds (0.5 Hz). Higher precision than DHT11 but slower and more expensive.

Pinout (3 or 4 pins depending on breakout): Pin 1 = VCC (3.3V or 5V), Pin 2 = DATA (connect to Arduino digital pin), Pin 3 = NC (not connected, on 4-pin version only), Pin 4 = GND. MANDATORY: A 10k ohm pull-up resistor must be connected between VCC and the DATA pin. Without this pull-up, the sensor will return NaN or zero readings. Many breakout boards include this resistor on-board.

```
Arduino Library: Use Adafruit DHT library. Include DHT.h. Initialize with: DHT dht(pin, DHT11) or DHT dht(pin, DHT22). Read with: float temp = dht.readTemperature(); float humid = dht.readHumidity(); Always check isnan(temp) before using the value.
```

3. Photoresistor (LDR / Light Dependent Resistor)

A photoresistor (also called LDR or CdS cell) changes its resistance based on ambient light intensity. In bright light: resistance drops to approximately 1k-10k ohms. In darkness: resistance increases to 200k-1M ohms. No polarity (not polarized, can be connected in either direction).

Voltage Divider Wiring for Arduino Analog Input: Connect one leg of the LDR to 5V. Connect the other leg to both an Arduino analog pin (A0) AND one leg of a fixed 10k ohm resistor. Connect the other leg of the 10k resistor to GND. This creates a voltage divider: analogRead(A0) returns 0-1023 proportional to light level. Higher reading = brighter light.

```
Reading: int lightLevel = analogRead(A0); Mapping: int brightness = map(lightLevel, 0, 1023, 0, 255); analogWrite(ledPin, brightness); This creates an automatic night-light that brightens the LED as ambient light decreases.
```

4. PIR Motion Sensor (HC-SR501)

Operating Voltage: 4.5V to 20V DC. Output: Digital HIGH (3.3V) when motion detected, LOW when idle. Detection Range: Up to 7 meters, 120 degree cone. Output Delay: Adjustable from 3 seconds to 300 seconds via onboard potentiometer. Sensitivity: Adjustable via second onboard potentiometer.

Pinout (3 pins): Pin 1 = VCC (5V to 12V), Pin 2 = OUTPUT (Digital signal, connect to Arduino digital pin), Pin 3 = GND. IMPORTANT: The PIR sensor requires a 30-60 second warm-up / calibration period after power-on before it provides reliable readings. During this time, ignore all output signals.

5. Potentiometer (Variable Resistor / Trimpot)

A potentiometer is a three-terminal variable resistor. Common values: 1k, 5k, 10k, 50k, 100k ohms. Rotation: 270 degrees (single-turn) or 10+ turns (multi-turn trimpot). The wiper (center pin) slides along a resistive element, producing a continuously variable output voltage between 0V and VCC.

Wiring: Pin 1 (left) = 5V or 3.3V. Pin 2 (center / wiper) = Arduino analog input (A0-A5). Pin 3 (right) = GND. Reading: int val = analogRead(A0); returns 0-1023. Common uses: Volume control, brightness dimming, servo position control, threshold adjustment.

6. IR Obstacle Avoidance Sensor (FC-51 / TCRT5000)

Operating Voltage: 3.3V to 5V. Output: Digital LOW when obstacle detected within range, HIGH when clear. Detection Range: 2cm to 30cm (adjustable via onboard potentiometer). Contains an IR LED transmitter and an IR photodiode receiver. The onboard comparator IC (LM393) produces a clean digital output.

Pinout: VCC = 3.3V or 5V, GND = Ground, OUT = Digital output to Arduino. This sensor is commonly used in line-following robots (detecting black tape on white surface) and obstacle-avoidance robots.

7. Soil Moisture Sensor

Operating Voltage: 3.3V to 5V. Output: Both analog (A0) and digital (D0) output pins. The analog output provides a value from 0 (completely wet) to 1023 (completely dry). The digital output triggers HIGH/LOW based on a threshold set by the onboard potentiometer. Probe material: Often corrosion-prone copper; capacitive soil moisture sensors (v1.2) are preferred for long-term use as they do not corrode.

Wiring: VCC to 5V, GND to GND, A0 to Arduino analog pin, D0 to Arduino digital pin (optional). Insert the two metal prongs into the soil. Wet soil conducts electricity better, lowering resistance and producing a lower analog reading.